NAKSHATRA GARG

Data Scientist | AI Engineer - LLM Inference, Generative AI (GenAI) & Voice AI

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with **3+ years** building production LLM and voice AI systems for enterprise clients. Own a **self-hosted open-weight LLM** on **vLLM** and **NVIDIA H200** GPUs serving **~2.5M voice-bot calls/day**, which replaced a third-party LLM API and cut inference costs **~40%** at **0.3 s p95** time-to-first-sentence. Hands-on across LLM inference optimization, guardrails, hybrid-retrieval RAG, LoRA fine-tuning, LLM evaluation and real-time ASR/TTS, from data and training to APIs, observability and multi-GPU production rollout.

SKILLS

LLM Serving & Inference: vLLM, self-hosted open-weight LLMs, streaming inference, latency optimization (TTFT, p95), CUDA graphs, multi-GPU deployment, GPU inference (NVIDIA H200, Blackwell, A100, T4), Ollama

Generative AI & NLP: Retrieval-Augmented Generation (RAG), hybrid retrieval (BM25 + dense vectors), reranking, embeddings (BGE-M3), LoRA/QLoRA fine-tuning, Hugging Face Transformers, prompt engineering, LLM evaluation (LLM-as-a-judge, human evaluation, precision@k), guardrails (PII redaction, content filtering), multimodal LLMs

Speech & Voice AI: ASR/STT, TTS, spoken language identification, voice cloning (RVC), SpeechBrain, Azure Speech Services, real-time voice bots

LLMOps & Infrastructure: Docker, HAProxy, Prometheus, Redis, model serving APIs, production monitoring, OCR

Programming & APIs: Python, FastAPI, Flask, WebSocket, REST APIs, SQL

Databases & Vector Search: Qdrant, PostgreSQL (pgvector), FAISS, ChromaDB, MySQL

Cloud: Azure (Azure ML, AI Services, Speech, Blob Storage, Compute), AWS (EC2, S3, Bedrock)

EXPERIENCE

Data Scientist / AI Associate | Rezo.AI | Noida, India | May 2024 - Present

- Owned a **self-hosted open-weight LLM** serving **~2.5M voice-bot calls/day** in production on **NVIDIA H200** GPUs with **vLLM**, replacing a third-party LLM API; built a **smart LLM router on HAProxy** to route requests, reduced inference costs **~40%**, eliminated external **data-exposure risk**, and matched or beat prior quality and latency.
- Introduced **Time-to-First-Sentence (TTFS)** as the primary voice latency metric instead of TTFT; achieved **50 ms TTFT**, **0.3 s p95 / 0.6 s p99 TTFS** and 1.5-2 s p95 full-response latency, tracked via real-time **Prometheus** dashboards.
- Designed a **multi-layer guardrail** and post-processing stack over the LLM serving layer: **PII redaction**, reasoning/thinking-token leak suppression, repetition control, harmful-content filtering and system-prompt instruction adherence, all within the production latency budget.
- Deployed a GPU-accelerated **voice language-identification** service (vLLM + open multimodal model) to production; cut audio-encode latency **35x** via **CUDA graph** capture and scaled to **100+ concurrent callers** at **sub-300 ms p95** with **98% accuracy**; owned APIs, observability and a multi-GPU rollout (A100 to NVIDIA Blackwell) that **doubled capacity**.
- Built production **hybrid-retrieval RAG** pipelines (BM25 + dense vector search in Qdrant + reranking) for LLM chatbots and voice bots across multiple enterprise clients, reaching **~80% precision@3**, reducing **hallucinations** and improving flow completion rates; backed by an embedding-extraction service with two-tier caching (Redis + Qdrant).
- Fine-tuned Qwen3.5-4B and Gemma-4B with **LoRA**, evaluated via **LLM-as-a-judge** and human review; currently own data and training pipelines for in-house **ASR and TTS** models.
- Developed a real-time **voice cloning** system on **RVC** with a custom headless, in-memory inference pipeline, reducing latency **20x** (from 20 s to ~1 s).
- Built and deployed an **OCR** system extracting data from Indian government ID documents (Aadhaar, PAN) for a leading Indian bank, powering production **KYC** and document verification.

Data Science Intern | Rezo.AI | Noida, India | Aug 2023 - Apr 2024

- Improved OCR accuracy **15%** by optimizing a multi-stage document processing pipeline (object detection to OCR), mitigating data drift through retraining and refined pre/post-processing.
- Designed **speech analytics** metrics for agent-customer calls (conversational flow, silence duration, interruption frequency, response latency) using Python pipelines over transcripts and audio metadata.

EDUCATION

M.Tech, Computer Science and Engineering (Minor in AI & ML) | VIT Vellore | 2024 | CGPA 8.9/10

B.Tech, Information Technology | College of Engineering Roorkee | 2022 | CGPA 8.4/10